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Laporan Sementara

Praktikum Jaringan Komputer

Firewall & NAT

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1 Pendahuluan

1.1 Latar Belakang

Dalam suatu topologi jaringan komputer, diperlukan mekanisme yang dapat mengatur lalu lintas data sekaligus menjaga keamanan jaringan dari akses yang tidak diinginkan. Untuk itu, diciptakanlah Firewall dan Network Address Translation (NAT). Firewall berfungsi untuk memfilter lalu lintas jaringan berdasarkan aturan tertentu, sedangkan NAT memungkinkan perangkat dalam jaringan lokal menggunakan alamat IP privat untuk terhubung ke internet melalui alamat IP publik. Pada praktikum ini, dilakukan konfigurasi Firewall dan NAT pada MikroTik untuk memahami cara kerja penerjemahan alamat IP serta penerapan aturan keamanan jaringan.

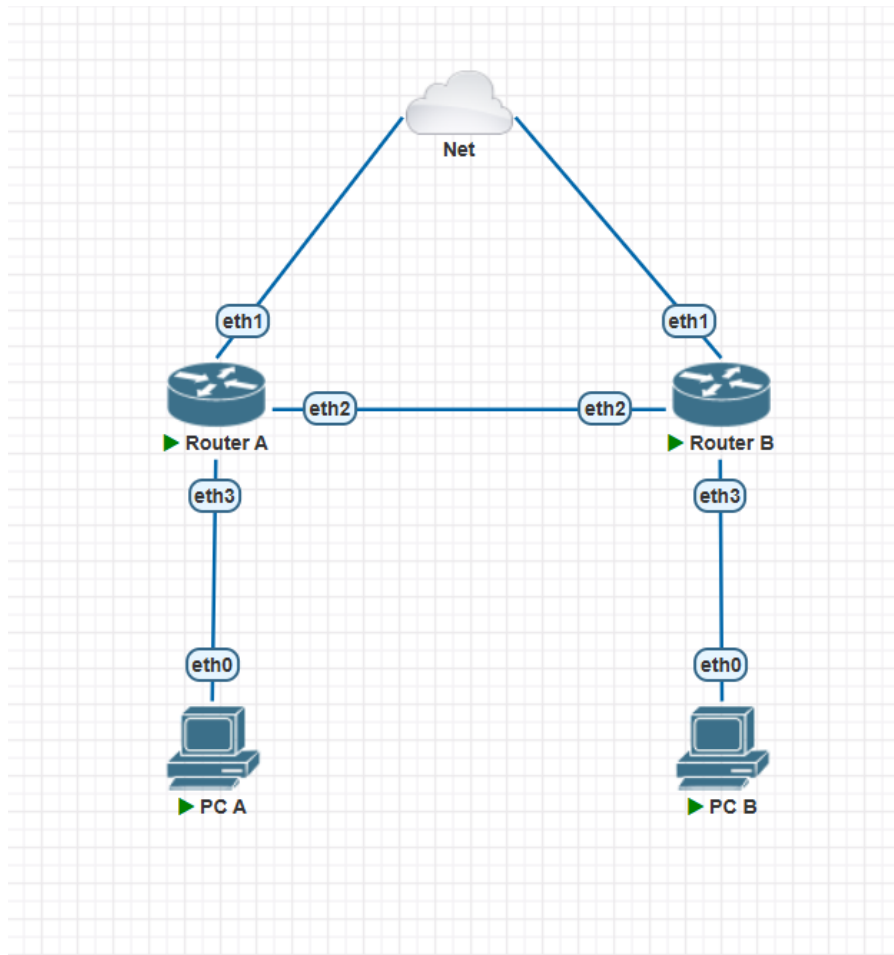
1.2 Dasar Teori

Firewall merupakan sistem keamanan jaringan yang berfungsi untuk menyaring lalu lintas data dari jaringan berdasarkan aturan yang telah ditentukan oleh administrator. Dengan adanya firewall, akses dapat dibatasi sehingga keamanan jaringan dapat lebih terjaga. Selain itu, terdapat teknologi Network Address Translation (NAT) yang digunakan untuk menerjemahkan alamat IP privat menjadi alamat IP publik atau sebaliknya. NAT memungkinkan beberapa perangkat dalam jaringan lokal untuk mengakses internet menggunakan satu alamat IP publik, sehingga lingkungan jaringan lebih aman karena alamat IP internal tidak langsung terekspos ke jaringan luar.

2 Tugas Pendahuluan

1. Topologi Jaringan





Gambar 1: Topologi

2. Konfigurasi Router A

```

Terminal
Router A x Router B x
[admin@MikroTik] > ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 10.10.10.1/30 10.10.10.0 ether2
1 10.10.10.5/30 10.10.10.4 ether3
2 D 10.4.76.28/20 10.4.64.0 ether1
[admin@MikroTik] > /ip dhcp-client
[admin@MikroTik] /ip dhcp-client> print
Flags: X - disabled, I - invalid, D - dynamic
# INTERFACE USE-PEER-DNS ADD-DEFAULT-ROUTE STATUS ADDRESS
0 ether1 yes yes bound 10.4.76.28/20
1 XI ether1 yes yes
[admin@MikroTik] /ip dhcp-client> /ip pool
[admin@MikroTik] /ip pool> print
# NAME RANGES
0 poolA 10.10.10.6
[admin@MikroTik] /ip pool> /ip dhcp-server network
[admin@MikroTik] /ip dhcp-server network> print
Flags: D - dynamic
# ADDRESS GATEWAY DNS-SERVER WINS-SERVER DOMAIN
0 0.0.0.0/0
1 10.10.10.4/30 10.10.10.5 8.8.8.8
[admin@MikroTik] /ip dhcp-server network> /ip dhcp-server

```

Gambar 2: Konfigurasi Router A

```

Terminal
Router A x Router B x
1 10.10.10.4/30 10.10.10.5 8.8.8.8
[admin@MikroTik] /ip dhcp-server network> /ip dhcp-server
[admin@MikroTik] /ip dhcp-server> print
Flags: D - dynamic, X - disabled, I - invalid
# NAME INTERFACE RELAY ADDRESS-POOL LEASE-TIME ADD-ARP
0 dhcpA ether3 poolA 10m
[admin@MikroTik] /ip dhcp-server> /ip firewall nat
[admin@MikroTik] /ip firewall nat> print
Flags: X - disabled, I - invalid, D - dynamic
0 chain=srcnat action=masquerade out-interface=ether1
[admin@MikroTik] /ip firewall nat> /ip route
[admin@MikroTik] /ip route> print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADS 0.0.0.0/0 10.4.64.1 1
1 ADC 10.4.64.0/20 10.4.76.28 ether1 0
2 ADC 10.10.10.0/30 10.10.10.1 ether2 0
3 ADC 10.10.10.4/30 10.10.10.5 ether3 0
[admin@MikroTik] /ip route> add dst-address=10.10.10.8/30 gateway=10.10.10.2
[admin@MikroTik] /ip route> print
Flags: X - disabled, A - active, D - dynamic,

```

Gambar 3: Konfigurasi Router A

```

Terminal
Router A x Router B x
0 chain=srcnat action=masquerade out-interface=ether1
[admin@MikroTik] /ip firewall nat> /ip route
[admin@MikroTik] /ip route> print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADS 0.0.0.0/0 10.4.64.1 1
1 ADC 10.4.64.0/20 10.4.76.28 ether1 0
2 ADC 10.10.10.0/30 10.10.10.1 ether2 0
3 ADC 10.10.10.4/30 10.10.10.5 ether3 0
[admin@MikroTik] /ip route> add dst-address=10.10.10.8/30 gateway=10.10.10.2
[admin@MikroTik] /ip route> print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADS 0.0.0.0/0 10.4.64.1 1
1 ADC 10.4.64.0/20 10.4.76.28 ether1 0
2 ADC 10.10.10.0/30 10.10.10.1 ether2 0
3 ADC 10.10.10.4/30 10.10.10.5 ether3 0
4 A S 10.10.10.8/30 10.10.10.2 1
[admin@MikroTik] /ip route>

```

Gambar 4: Konfigurasi Router A

3. Konfigurasi Router B

```

Terminal
Router B x
[admin@MikroTik] > ip dhcp-server print
Flags: D - dynamic, X - disabled, I - invalid
# NAME INTERFACE RELAY ADDRESS-POOL LEASE-TIME ADD-ARP
0 dhcpB ether3 poolB 10m
[admin@MikroTik] > ip firewall nat print
Flags: X - disabled, I - invalid, D - dynamic
[admin@MikroTik] > /ip firewall nat
[admin@MikroTik] /ip firewall nat> add chain=srcnat out-interface=ether1 action=masquerade
[admin@MikroTik] /ip firewall nat> print
Flags: X - disabled, I - invalid, D - dynamic
0 chain=srcnat action=masquerade out-interface=ether1
[admin@MikroTik] /ip firewall nat> /ip route
[admin@MikroTik] /ip route> add dst-address=10.10.10.4/30 gateway= 10.10.10.1
[admin@MikroTik] /ip route> print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADS 0.0.0.0/0 10.4.64.1 1
1 ADC 10.4.64.0/20 10.4.76.29 ether1 0
2 ADC 10.10.10.0/30 10.10.10.2 ether2 0
3 A S 10.10.10.4/30 10.10.10.1 1
4 ADC 10.10.10.8/30 10.10.10.9 ether3 0

```

Gambar 5: Konfigurasi Router B

```

Terminal
Router B x
[admin@MikroTik] /ip dhcp-server> /
[admin@MikroTik] > ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 10.10.10.2/30 10.10.10.0 ether2
1 10.10.10.9/30 10.10.10.8 ether3
2 D 10.4.76.29/20 10.4.64.0 ether1
[admin@MikroTik] > /ip dhcp-client print
Flags: X - disabled, I - invalid, D - dynamic
# INTERFACE USE-PEER-DNS ADD-DEFAULT-ROUTE STATUS ADDRESS
0 ether1 yes yes bound 10.4.76.29/20
1 XI ether1 yes yes
[admin@MikroTik] > ip pool print
# NAME RANGES
0 poolB 10.10.10.10
[admin@MikroTik] > ip dhcp-server network print
Flags: D - dynamic
# ADDRESS GATEWAY DNS-SERVER WINS-SERVER DOMAIN
0 10.10.10.8/30 10.10.10.9 8.8.8.8
[admin@MikroTik] > ip dhcp-server print
Flags: D - dynamic, X - disabled, I - invalid
# NAME INTERFACE RELAY ADDRESS-POOL LEASE-TIME ADD-ARP
0 dhcpB ether3 poolB 10m

```

Gambar 6: Konfigurasi Router B

4. Konfigurasi PC A

```

Terminal
Router B x PCA x PC B x
VPCS> ping 10.10.10.10

84 bytes from 10.10.10.10 icmp_seq=1 ttl=62 time=2.268 ms
84 bytes from 10.10.10.10 icmp_seq=2 ttl=62 time=1.337 ms
84 bytes from 10.10.10.10 icmp_seq=3 ttl=62 time=1.327 ms
84 bytes from 10.10.10.10 icmp_seq=4 ttl=62 time=1.353 ms
84 bytes from 10.10.10.10 icmp_seq=5 ttl=62 time=1.439 ms

VPCS> show ip
NAME      : VPCS[1]
IP/MASK   : 10.10.10.6/30
GATEWAY   : 10.10.10.5
DNS       : 8.8.8.8
DHCP SERVER : 10.10.10.5
DHCP LEASE : 551, 600/300/525
MAC       : 00:50:79:66:68:33
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU       : 1500
VPCS>

```

Gambar 7: Konfigurasi PC A

5. Konfigurasi PC B

```

Terminal
Router B x PCA x PC B x
VPCS> ip addr
Invalid address

VPCS> ip address print
Invalid options

VPCS> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=112 time=24.005 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=112 time=24.274 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=112 time=22.897 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=112 time=23.145 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=112 time=22.666 ms

VPCS> ping 10.10.10.6

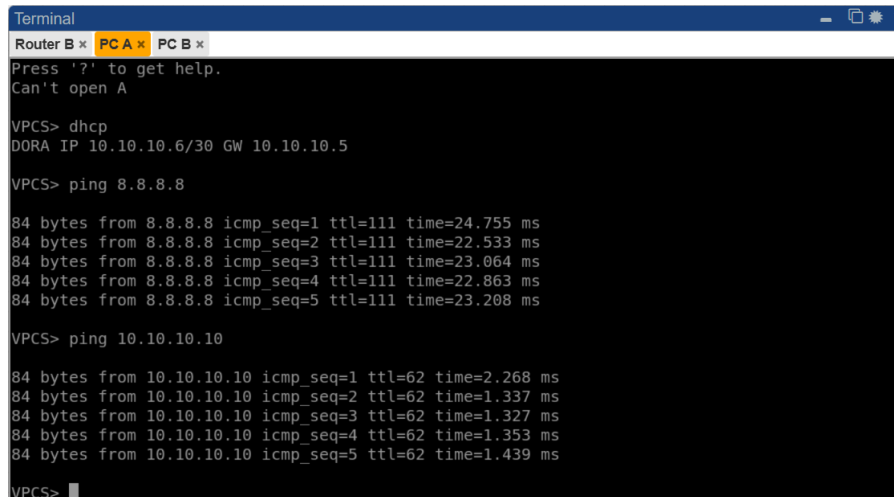
84 bytes from 10.10.10.6 icmp_seq=1 ttl=62 time=1.097 ms
84 bytes from 10.10.10.6 icmp_seq=2 ttl=62 time=2.435 ms
84 bytes from 10.10.10.6 icmp_seq=3 ttl=62 time=0.813 ms
84 bytes from 10.10.10.6 icmp_seq=4 ttl=62 time=1.186 ms
84 bytes from 10.10.10.6 icmp_seq=5 ttl=62 time=1.614 ms

VPCS>

```

Gambar 8: Konfigurasi PC B

6. Test Koneksi PC A



A terminal window titled 'Terminal' with tabs for 'Router B', 'PC A', and 'PC B'. The 'PC A' tab is active. The terminal shows the following commands and output:

```
Router B x PC A x PC B x
Press '?' to get help.
Can't open A

VPCS> dhcp
DORA IP 10.10.10.6/30 GW 10.10.10.5

VPCS> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=111 time=24.755 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=111 time=22.533 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=111 time=23.064 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=111 time=22.863 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=111 time=23.208 ms

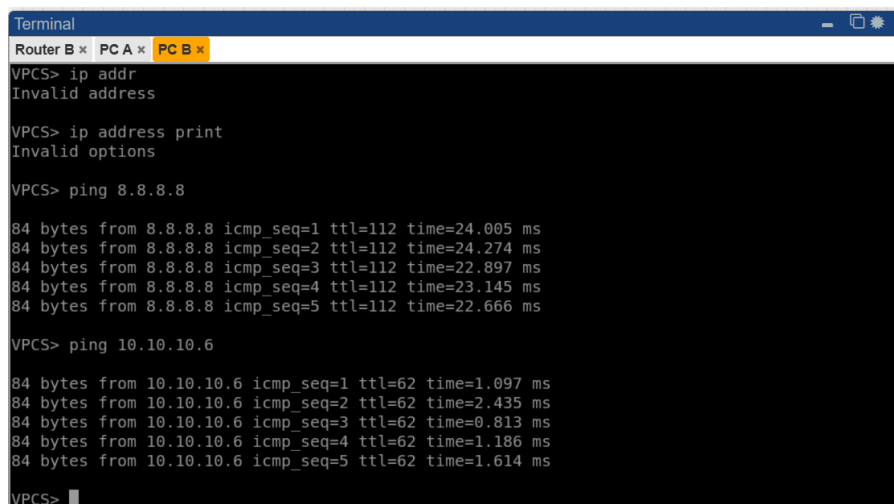
VPCS> ping 10.10.10.10

84 bytes from 10.10.10.10 icmp_seq=1 ttl=62 time=2.268 ms
84 bytes from 10.10.10.10 icmp_seq=2 ttl=62 time=1.337 ms
84 bytes from 10.10.10.10 icmp_seq=3 ttl=62 time=1.327 ms
84 bytes from 10.10.10.10 icmp_seq=4 ttl=62 time=1.353 ms
84 bytes from 10.10.10.10 icmp_seq=5 ttl=62 time=1.439 ms

VPCS>
```

Gambar 9: Test Koneksi PC A

7. Test Koneksi PC B



A terminal window titled 'Terminal' with tabs for 'Router B', 'PC A', and 'PC B'. The 'PC B' tab is active. The terminal shows the following commands and output:

```
Router B x PC A x PC B x
VPCS> ip addr
Invalid address

VPCS> ip address print
Invalid options

VPCS> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=112 time=24.005 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=112 time=24.274 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=112 time=22.897 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=112 time=23.145 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=112 time=22.666 ms

VPCS> ping 10.10.10.6

84 bytes from 10.10.10.6 icmp_seq=1 ttl=62 time=1.097 ms
84 bytes from 10.10.10.6 icmp_seq=2 ttl=62 time=2.435 ms
84 bytes from 10.10.10.6 icmp_seq=3 ttl=62 time=0.813 ms
84 bytes from 10.10.10.6 icmp_seq=4 ttl=62 time=1.186 ms
84 bytes from 10.10.10.6 icmp_seq=5 ttl=62 time=1.614 ms

VPCS>
```

Gambar 10: Test Koneksi PC B